

BankAccount.java

```
public abstract class BankAccount {

    // Encapsulation - private data members
    private String accountNumber;
    private String accountHolderName;
    private double balance;

    public BankAccount(String accountNumber, String accountHolderName, double
balance) {
        this.accountNumber = accountNumber;
        this.accountHolderName = accountHolderName;
        this.balance = balance;
    }

    public String getAccountNumber() {
        return accountNumber;
    }

    public void setAccountNumber(String accountNumber) {
        this.accountNumber = accountNumber;
    }

    public String getAccountHolderName() {
        return accountHolderName;
    }

    public void setAccountHolderName(String accountHolderName) {
        this.accountHolderName = accountHolderName;
    }

    public double getBalance() {
        return balance;
    }

    public void setBalance(double balance) {
        this.balance = balance;
    }

    // Deposit method
    public void deposit(double amount) {
        balance += amount;
        System.out.println("Deposited: ₹" + amount);
    }

    // Display account details
    public void displayDetails() {
        System.out.println("Account Number    : " + accountNumber);
        System.out.println("Account Holder    : " + accountHolderName);
    }
}
```

```
        System.out.println("Balance          : ₹" + balance);  
    }  
  
    // Abstract method - Abstraction  
    public abstract double calculateInterest();  
}
```

```
// CurrentAccount.java
```

```
public class CurrentAccount extends BankAccount {

    private double interestRate = 2.0; // 2% interest

    // Constructor
    public CurrentAccount(String accountNumber, String accountHolderName, double
balance) {
        super(accountNumber, accountHolderName, balance);
    }

    // Implementing abstract method
    @Override
    public double calculateInterest() {
        return (getBalance() * interestRate) / 100;
    }

    // Display details
    @Override
    public void displayDetails() {
        System.out.println("--- Current Account ---");
        super.displayDetails();
        System.out.println("Interest Rate      : " + interestRate + "%");
        System.out.println("Interest Earned   : ₹" + calculateInterest());
    }
}
```

```
// Main.java
```

```
public class Main {  
    public static void main(String[] args) {  
  
        // Creating SavingsAccount object  
        SavingsAccount savings = new SavingsAccount("SA1001", "Aarav Sharma",  
50000);  
        savings.deposit(10000);  
        savings.displayDetails();  
  
        System.out.println();  
  
        // Creating CurrentAccount object  
        CurrentAccount current = new CurrentAccount("CA2001", "Priya Mehta",  
100000);  
        current.deposit(25000);  
        current.displayDetails();  
    }  
}
```

```
// SavingsAccount.java
```

```
public class SavingsAccount extends BankAccount {

    private double interestRate = 4.0; // 4% interest

    // Constructor
    public SavingsAccount(String accountNumber, String accountHolderName, double
balance) {
        super(accountNumber, accountHolderName, balance);
    }

    // Implementing abstract method
    @Override
    public double calculateInterest() {
        return (getBalance() * interestRate) / 100;
    }

    // Display details
    @Override
    public void displayDetails() {
        System.out.println("--- Savings Account ---");
        super.displayDetails();
        System.out.println("Interest Rate      : " + interestRate + "%");
        System.out.println("Interest Earned   : ₹" + calculateInterest());
    }
}
```

DemoApplication.java

```
package com.example.demo;
```

```
import org.springframework.boot.SpringApplication;
```

```
import org.springframework.boot.autoconfigure.SpringBootApplication;
```

```
@SpringBootApplication
```

```
public class DemoApplication {
```

```
    public static void main(String[] args) {
```

```
        SpringApplication.run(DemoApplication.class, args);
```

```
    }
```

```
}
```

Student.java

```
package com.example.demo;
```

```
public class Student {
```

```
    private int id;  
    private String name;  
    private String course;
```

```
    // Constructor
```

```
    public Student(int id, String name, String course) {  
        this.id = id;  
        this.name = name;  
        this.course = course;  
    }
```

```
    // Getters and Setters
```

```
    public int getId() { return id; }  
    public void setId(int id) { this.id = id; }
```

```
    public String getName() { return name; }  
    public void setName(String name) { this.name = name; }
```

```
    public String getCourse() { return course; }  
    public void setCourse(String course) { this.course = course; }
```

```
}
```

Employee.java

```
package com.example.employee;
```

```
import jakarta.persistence.*;
```

```
@Entity
```

```
public class Employee {
```

```
    @Id
```

```
    @GeneratedValue(strategy = GenerationType.IDENTITY)
```

```
    private int id;
```

```
    private String name;
```

```
    private String department;
```

```
    private double salary;
```

```
    // Constructors
```

```
    public Employee() {}
```

```
    public Employee(String name, String department, double salary) {
```

```
        this.name = name;
```

```
        this.department = department;
```

```
        this.salary = salary;
```

```
    }
```

```
    // Getters and Setters
```

```
    public int getId() { return id; }
```

```
    public void setId(int id) { this.id = id; }
```

```
    public String getName() { return name; }
```

```
    public void setName(String name) { this.name = name; }
```

```
    public String getDepartment() { return department; }
```

```
    public void setDepartment(String department) { this.department = department; }
```

```
    public double getSalary() { return salary; }
```

```
    public void setSalary(double salary) { this.salary = salary; }
```

```
}
```


EmployeeApplication.java

```
package com.example.employee;
```

```
import org.springframework.boot.SpringApplication;
```

```
import org.springframework.boot.autoconfigure.SpringBootApplication;
```

```
@SpringBootApplication
```

```
public class EmployeeApplication {
```

```
    public static void main(String[] args) {
```

```
        SpringApplication.run(EmployeeApplication.class, args);
```

```
    }
```

```
}
```

EmployeeController.java

```
package com.example.employee;
```

```
import org.springframework.beans.factory.annotation.Autowired;  
import org.springframework.web.bind.annotation.*;
```

```
import java.util.List;  
import java.util.Optional;
```

```
@RestController
```

```
@RequestMapping("/api/employees")
```

```
public class EmployeeController {
```

```
    @Autowired
```

```
    private EmployeeService service;
```

```
    // POST - Add new employee
```

```
    @PostMapping
```

```
    public Employee addEmployee(@RequestBody Employee employee) {  
        return service.addEmployee(employee);  
    }
```

```
    // GET - Get all employees
```

```
    @GetMapping
```

```
    public List<Employee> getAllEmployees() {  
        return service.getAllEmployees();  
    }
```

```
    // GET - Get employee by ID
```

```
    @GetMapping("/{id}")
```

```
    public Optional<Employee> getEmployeeById(@PathVariable int id) {  
        return service.getEmployeeById(id);  
    }
```

```
    // PUT - Update employee
```

```
    @PutMapping("/{id}")
```

```
    public Employee updateEmployee(@PathVariable int id, @RequestBody Employee  
employee) {  
        return service.updateEmployee(id, employee);  
    }
```

```
    // DELETE - Delete employee
```

```
    @DeleteMapping("/{id}")
```

```
    public String deleteEmployee(@PathVariable int id) {  
        return service.deleteEmployee(id);  
    }
```

```
}
```

EmployeeRepository.java

```
package com.example.employee;

import org.springframework.data.jpa.repository.JpaRepository;
import org.springframework.stereotype.Repository;

@Repository
public interface EmployeeRepository extends JpaRepository<Employee, Integer> {
    // JpaRepository provides all CRUD methods automatically
}
```

EmployeeService.java

```
package com.example.employee;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Service;

import java.util.List;
import java.util.Optional;

@Service
public class EmployeeService {

    @Autowired
    private EmployeeRepository repository;

    // Add employee
    public Employee addEmployee(Employee employee) {
        return repository.save(employee);
    }

    // Get all employees
    public List<Employee> getAllEmployees() {
        return repository.findAll();
    }

    // Get employee by ID
    public Optional<Employee> getEmployeeById(int id) {
        return repository.findById(id);
    }

    // Update employee
    public Employee updateEmployee(int id, Employee updatedEmployee) {
        Employee employee = repository.findById(id).orElseThrow();
        employee.setName(updatedEmployee.getName());
        employee.setDepartment(updatedEmployee.getDepartment());
        employee.setSalary(updatedEmployee.getSalary());
        return repository.save(employee);
    }

    // Delete employee
    public String deleteEmployee(int id) {
        repository.deleteById(id);
        return "Employee with ID " + id + " deleted successfully.";
    }
}
```